

Problem

The integral $\int_{-\infty}^{\infty} \frac{10\delta(\omega - 1)}{4 + \omega^2} d\omega$ gives:

- (a) 0
- (b) 2
- (c) 2.5
- (d) ∞

Step-by-step solution

Step 1 of 1

Consider the following integral.

$$\int_{-\infty}^{\infty} \frac{10\delta(\omega - 1)}{4 + \omega^2} d\omega$$

The following is the sampling or sifting property of the impulse function.

$$\int_a^b f(t) \delta(t - t_0) dt = f(t_0)$$

Use the sampling or sifting property of impulse function to evaluate the integral.

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{10\delta(\omega - 1)}{4 + \omega^2} d\omega &= \int_{-\infty}^{\infty} \left(\frac{10}{4 + \omega^2} \right) \delta(\omega - 1) d\omega \\ &= \left(\frac{10}{4 + \omega^2} \right)_{\omega=1} \\ &= \frac{10}{4 + 1} \\ &= 2 \end{aligned}$$

Thus, the integral $\int_{-\infty}^{\infty} \frac{10\delta(\omega - 1)}{4 + \omega^2} d\omega$ gives and the correct choice is .